

Exact Computation of Transfer Entropy with Path Weight Sampling

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The ability to quantify the directional flow of information is vital to understanding natural systems and designing engineered information-processing systems. A widely used measure to quantify this information flow is the transfer entropy. However, until now, this quantity could only be obtained in dynamical models using approximations that are typically uncontrolled. Here we introduce a computational algorithm called “transfer entropy-path weight sampling” (TE-PWS), which makes it possible, for the first time, to quantify the transfer entropy and its variants exactly for any stochastic model, including those with multiple hidden variables, nonlinearity, transient conditions, and feedback. By leveraging techniques from polymer and path sampling, TE-PWS efficiently computes the transfer entropy as a Monte Carlo average over signal trajectory space. We use our exact technique to demonstrate that commonly used approximate methods to compute transfer entropies incur large systematic errors and high computational costs. As an application, we use TE-PWS in linear and nonlinear systems to reveal how transfer entropy can overcome naive applications of the data processing inequality in the presence of feedback.

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Information transfer between noisy signals underlies the functionality of diverse natural and man-made networks such as those in biochemical signaling, neuroscience, ecology, wireless communication, and finance. Information theory has so far provided a useful framework for quantifying information transmission. In the presence of feedback loops in the network, information travels in both directions between the input and the output. An information-theoretic measure that can quantify the information transfer separately in either direction is the transfer entropy [1,2]. Transfer entropy and its variants, such as directed information, conditional transfer entropy, or filtered transfer entropy, have been widely used to gain knowledge about the connectivity of a network [3], infer causal relations from experiments [4,5], establish fundamental bounds on network performance [6], and estimate the minimal physical work required for a computation [7,8]. Hence, for a wide range of problems, it is vital to be able to accurately quantify transfer entropies.

However, there are currently no exact methods to compute the transfer entropy in a general many-variable dynamical model. The transfer entropy depends on probability distributions of signal trajectories. Estimating these distributions by binning the experimental or simulated trajectories in histograms is not feasible, as the dimensionality of the trajectory space scales exponentially with signal duration [9,10]. As a result, different approximations are currently being used: either the full history dependence of the transfer entropy is truncated [11], an arbitrary distance

metric in trajectory space is chosen for clustering the trajectories [12], or a linear or low-order moment-closure approximation is employed [13–15]. These approximations can result in uncontrolled errors in complex, nonlinear, many-variable systems [16,17]. In the absence of an exact method, the magnitude of these errors remains unclear.

Here, we fill this gap by introducing TE-PWS, a numerical algorithm to estimate transfer entropies exactly for any stochastic model, including diffusive and jump processes. The estimate is exact, i.e., it is an unbiased statistical estimate of the transfer entropy. TE-PWS can therefore provide *ground truth* results for any given model. TE-PWS builds on the recently developed PWS algorithm for computing the mutual information between trajectories [18]. TE-PWS exploits the idea that path likelihoods can be obtained analytically from the Langevin or master equation, from which the transfer entropy is then computed via Monte Carlo averaging in trajectory space. Additionally, long trajectories are sampled with an importance sampling scheme [19], solving the problem of exponential scaling of the computational cost with trajectory duration. We first show that TE-PWS reproduces analytical results when available. We then apply TE-PWS to compute the transfer entropy in a three-variable motif, in the presence of feedback, for both linear and nonlinear systems, yielding novel insights on how information feedback can amplify information transfer. Specifically, the transfer entropy from an input to an output node can overcome a naive application of the data processing inequality even when the mutual information obeys one. Finally, we use TE-PWS to show that the most widely used approximate methods for the computation of transfer entropy—the Gaussian framework,

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the Kraskov-Stögbauer-Grassberger (KSG) algorithm, and single-step truncation—all exhibit significant systematic errors. The comparison also reveals that TE-PWS is computationally efficient.

Transfer entropy—The TE-PWS algorithm has the same key steps for all stochastic models. We therefore describe TE-PWS for diffusive processes in the main text and for jump processes in the End Matter (EM). Consider a d -dimensional diffusive process $\mathbf{X}(t)$ modeled as a function of time t by a Langevin equation,

$$\dot{\mathbf{X}}(t) = \mathbf{F}(t) + \boldsymbol{\xi}(t), \quad (1)$$

with $\mathbf{F}(t)$ a general drift, and $\boldsymbol{\xi}(t)$ a d -dimensional Gaussian white noise with a diffusion constant matrix $\mathbf{D} = [D_{ij}]$ such that $\langle \xi_i(t) \xi_j(t') \rangle = 2D_{ij} \delta(t - t')$. The drift may depend on the entire past history as well as on time. The transfer entropy from X_i to X_j over N time steps of durations δt each is defined as [1]

$$\mathcal{T}_{X_i \rightarrow X_j} = \sum_{k=0}^{N-1} I(X_j(k+1); X_{i,[0,k]} | X_{j,[0,k]}) \quad (2)$$

$$\equiv \sum_{k=0}^{N-1} \mathcal{T}_{X_i \rightarrow X_j}^{[k]}, \quad (3)$$

where the index k goes over individual time steps, $X_j(k)$ denotes X_j after k time steps, $X_{j,[0,k]}$ denotes the trajectory over the first k time steps, $\mathcal{T}_{X_i \rightarrow X_j}^{[k]}$ denotes the k th term in the sum, and $I(A; B)$ denotes the mutual information between two random variables A and B . $\mathcal{T}_{X_i \rightarrow X_j}$ measures the information transferred from the past trajectory of X_i to the new updates of X_j at every time step, given the past trajectory of X_j is already known. In case the dynamics is in steady state, we will also talk about the transfer entropy rate, $\dot{\mathcal{T}}_{X_i \rightarrow X_j} = \lim_{N \rightarrow \infty} \mathcal{T}_{X_i \rightarrow X_j} / (N \delta t)$.

We can rewrite the transfer entropy equivalently as

$$\mathcal{T}_{X_i \rightarrow X_j} = \sum_{k=0}^{N-1} \mathcal{T}_{X_i \rightarrow X_j}^{[k]} = \sum_k H(X_j(k+1) | X_{j,[0,k]}) - H(X_j(k+1) | X_{i,[0,k]}, X_{j,[0,k]}) \quad (4)$$

$$= \sum_k \left\langle \ln \frac{P(X_j(k+1) | X_{i,[0,k]}, X_{j,[0,k]})}{P(X_j(k+1) | X_{j,[0,k]})} \right\rangle, \quad (5)$$

where $H(A)$ denotes the Shannon entropy associated with the probability distribution $P(A)$ of A , and the angular brackets denote an average over the joint probability $P(X_{i,[0,N]}, X_{j,[0,N]})$. Equation (4) shows that transfer entropy quantifies the additional information in $X_j(k+1)$ that arrives from $X_{i,[0,k]}$ beyond that which is already present

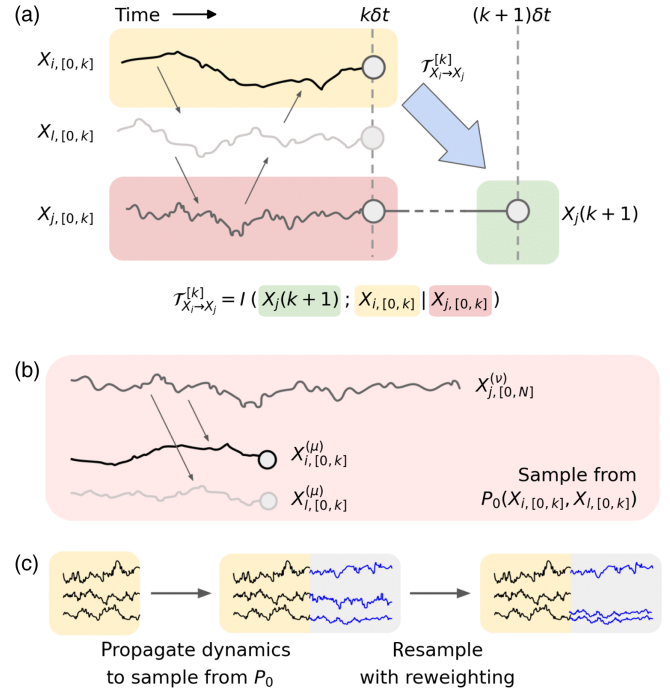


FIG. 1. (a) Schematic representation of the increase in transfer entropy from variable X_i to X_j at the $(k+1)$ th time step. Other variables X_l may mediate information transfer even in the absence of a direct coupling from X_i to X_j . (b) Propagation of reference dynamics for X_i and X_l such that it is commensurate with the given frozen $X_j^{(\nu)}$ trajectory. (c) In the RR scheme, trajectories sampled from a reference distribution $P_0(X_{i,[0,k]}, X_{l,[0,k]})$ are resampled periodically to turn them into the desired conditional distribution $P(X_{i,[0,k]}, X_{l,[0,k]} | X_j^{(\nu)})$.

in the past trajectory $X_{j,[0,k]}$. This occurs either through direct causal action, or through a third variable X_l , schematically demonstrated in Fig. 1(a). If X_i does not affect the dynamics of X_j , this additional information would be zero. In general, every $\mathcal{T}_{X_i \rightarrow X_j}^{[k]}$ is a mutual information and hence nonnegative.

For calculating the transfer entropy using Eq. (5) we develop TE-PWS. Following PWS [18], the central idea is that trajectory likelihoods in the full d -dimensional space are analytically available on the fly, and that trajectory averages can be computed in a Monte Carlo fashion. First, the average in Eq. (5) is computed as

$$\mathcal{T}_{X_i \rightarrow X_j} = \frac{1}{M_1} \sum_{\nu} \sum_k \ln \frac{P(X_j^{(\nu)}(k+1) | X_{i,[0,k]}, X_{j,[0,k]})}{P(X_j^{(\nu)}(k+1) | X_{j,[0,k]})}, \quad (6)$$

where the index ν sums over M_1 pairs of trajectories of X_i and X_j sampled from the joint probability distribution $P(X_{i,[0,N]}, X_{j,[0,N]})$. The superscript (ν) in $X_i^{(\nu)}$ refers to the ν th statistical realization of X_i . For each pair of trajectories,

the probabilities in the numerator and the denominator of Eq. (6) are not analytically available, but what is indeed available is the full joint probability $P(\mathbf{X}_{[0,N]}^{(\nu)})$ as the exponential of the Onsager-Machlup action [20,21]. We thus need to marginalize over all degrees of freedom other than X_i and X_j , denoted henceforth collectively as X_l . We illustrate this procedure first for the denominator in Eq. (6). It is obtained as $P(X_{j,[0,k+1]}^{(\nu)})/P(X_{j,[0,k]}^{(\nu)})$, where $P(X_{j,[0,k]}^{(\nu)})$ is obtained via marginalization,

$$P(X_{j,[0,k]}^{(\nu)}) = \int \int D[X_{i,[0,k]}] D[X_{l,[0,k]}] P(X_{i,[0,k]}, X_{j,[0,k]}^{(\nu)}, X_{l,[0,k]}). \quad (7)$$

For performing this average in a Monte Carlo fashion [19], we sample from a reference distribution, $P_0(X_{i,[0,k]}, X_{l,[0,k]})$, and correct the resultant bias by dividing by P_0 ,

$$P(X_{j,[0,k]}^{(\nu)}) = \frac{1}{M_2} \sum_{\mu} \frac{P(X_{i,[0,k]}^{(\mu)}, X_{j,[0,k]}^{(\nu)}, X_{l,[0,k]}^{(\mu)})}{P_0(X_{i,[0,k]}^{(\mu)}, X_{l,[0,k]}^{(\mu)})}, \quad (8)$$

where the index μ sums over M_2 trajectories sampled from P_0 . What is the best choice for P_0 ? The ideal choice would be the conditional distribution $P(X_{i,[0,k]}, X_{l,[0,k]} | X_{j,[0,k]}^{(\nu)})$, as it makes the summand in Eq. (8) equal for all μ , such that the variance of the estimate of $P(X_{j,[0,k]}^{(\nu)})$ is zero. However, this conditional distribution is not known *a priori*. We therefore generate X_i and X_l trajectories in the frozen field of $X_{j,[0,N]}^{(\nu)}$ [Fig. 1(b)] resulting in a distribution P_0 that is known analytically and is expected to be close to this conditional distribution. To exactly compensate for the remaining deviations of P_0 from the desired conditional distribution, we employ, in the spirit of Rosenbluth-Rosenbluth (RR) PWS [18], a reweighing of the $X_i^{(\mu)}$ and $X_l^{(\mu)}$ trajectories on the fly with weights proportional to the ratio of the two distributions. We resample the trajectories with these weights after every δt time, meaning that we sample M_2 trajectories with replacement from the weighted trajectory ensemble [Fig. 1(c)]. This procedure exactly generates $X_i^{(\mu)}$ and $X_l^{(\mu)}$ trajectories according to the desired conditional distribution $P(X_{i,[0,k]}, X_{l,[0,k]} | X_{j,[0,k]}^{(\nu)})$ (see EM and Supplemental Material (SM) [22], Sec. SM-C, which include Ref. [23] and SM-D, which include Ref. [24]).

Returning now to the numerator in Eq. (6), we note that it is an average over another conditional distribution,

$$\begin{aligned} & P(X_j^{(\nu)}(k+1) | X_{i,[0,k]}^{(\nu)}, X_{j,[0,k]}^{(\nu)}) \\ &= \int D[X_{l,[0,k]}] P(X_j^{(\nu)}(k+1) | X_{i,[0,k]}^{(\nu)}, X_{j,[0,k]}^{(\nu)}, X_{l,[0,k]}) \\ &\quad \times P(X_{l,[0,k]} | X_{i,[0,k]}^{(\nu)}, X_{j,[0,k]}^{(\nu)}). \end{aligned} \quad (9)$$

The first probability in the integral, which is the transition probability of X_j in the full d -dimensional space, is analytically available. Additionally, similar to the procedure for the denominator in Eq. (6), samples from $P(X_{l,[0,k]} | X_{i,[0,k]}^{(\nu)}, X_{j,[0,k]}^{(\nu)})$ are also available by sampling X_l trajectories first from a $P_0(X_{l,[0,k]})$, and then applying the RR scheme. Thus the numerator in Eq. (6) can also be evaluated in a Monte Carlo fashion (see EM).

To summarize, for each of the M_1 pairs of $X_{i,[0,N]}$ and $X_{j,[0,N]}$ trajectories, we simulate a joint ensemble of M_2 new X_i and X_l trajectories to estimate the denominator in the logarithm in Eq. (6), and a separate ensemble of M_2 new X_l trajectories to estimate the numerator (Fig. 1). The trajectories in each ensemble are resampled on the fly with the RR scheme, giving both the numerator and denominator in Eq. (6) as Monte Carlo averages. The computational cost thus scales as $2M_1M_2$. The transfer entropy estimate is unbiased [18], and the statistical accuracy can be arbitrarily improved by increasing M_1 and M_2 . A pseudocode for the algorithm is available in SM [22] Sec. SM-B. Aside from Schreiber's transfer entropy [1], other trajectory-based metrics of directional information transfer such as directed information [25], conditional transfer entropy [26], and filtered transfer entropy [27], also can be derived from conditional distributions of trajectories. Hence, TE-PWS can be used to compute all such metrics at similar cost, as shown in SM [22] Sec. SM-G, which include Refs. [28,29].

We demonstrate the validity of the method by analyzing two examples of an Ornstein-Uhlenbeck (OU) process with feedback for which transfer entropy rates are exactly available since the trajectories are Gaussian-distributed. Equation (1) represents an OU process when it is linear with $\mathbf{F} = -\mathbf{a}\mathbf{X}$, where $\mathbf{a}$ is a spring constant matrix. The first example is a two-variable OU process (model A) for which the transfer entropy rates are analytically available (see EM). Additionally, since the trajectories are Gaussian-distributed, each term in Eq. (3) can be exactly computed by estimating the covariance of trajectories, a commonly used approach called the Gaussian framework for the transfer entropy [13,14] (see SM [22] Sec. SM-H). On comparing these values with TE-PWS in Fig. 2(a), we find that TE-PWS gives accurate and unbiased estimates of $\dot{\mathcal{T}}_{X_1 \rightarrow X_2}^{[k]} \equiv \mathcal{T}_{X_1 \rightarrow X_2}^{[k]}/\delta t$ for every k , converging to $\dot{\mathcal{T}}_{X_1 \rightarrow X_2}$ for large k . A second benchmark is a three-variable model (model B) for describing the stochastic dynamics of gene expression and the growth rate of bacteria [30]. The model is nonbipartite, meaning that the diffusion constant matrix

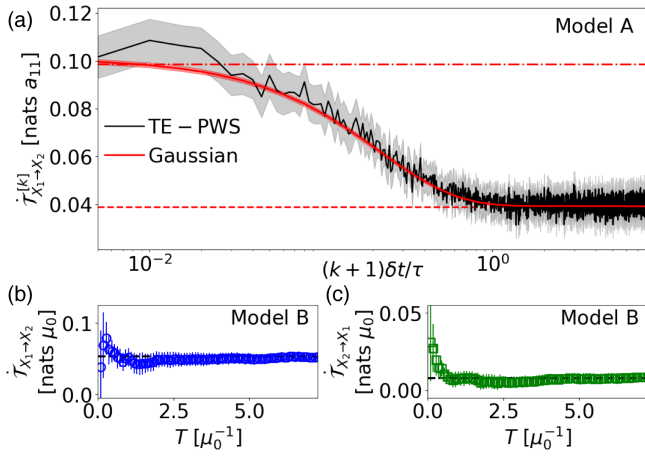


FIG. 2. Convergence of transfer entropy rate estimates in linear models. (a) shows $\dot{T}_{X_1 \rightarrow X_2}^{[k]} \equiv \mathcal{T}_{X_1 \rightarrow X_2}^{[k]}/\delta t$ in the two-variable OU process (model A) with feedback as a function of the history length $k + 1$. The black line is from TE-PWS, the solid red line is the exact result from the Gaussian framework, and the red dashed-dot and dashed lines are analytical results for $\dot{T}_{X_1 \rightarrow X_2}^{[k=0]}$ (single-step) and $\dot{T}_{X_1 \rightarrow X_2}$, respectively. (b) and (c) show transfer entropy rates in the linear stochastic gene expression model (model B) [30]. Symbols are from TE-PWS and dashed lines are semianalytical results.

is nondiagonal and the mutual information rates are not finite. Yet transfer entropy rates are finite and semianalytically available [27]. As shown in Figs. 2(b) and 2(c), the steady-state transfer entropy rate from TE-PWS converges to the known values. Details about the models are provided in SM [22] Sec. SM-A, which include Refs. [31,32].

Data processing inequality—We demonstrate the utility of TE-PWS by applying it to a three-node motif to study whether transfer entropies obey the data processing inequality (DPI). Unidirectional flow of information between different nodes in a network leads to a DPI for the mutual information [33]. For a general three-variable process, if the flow of information is $X_1 \rightarrow X_2 \rightarrow X_3$, i.e., without feedback, $I(X_{1,[0,N]}; X_{3,[0,N]} | X_{2,[0,N]}) = 0$; here, the right arrows denote flow of information as mediated either via activation or repression. This leads by the chain rule to $I(X_{1,[0,N]}; X_{3,[0,N]}) \leq I(X_{1,[0,N]}; X_{2,[0,N]})$ [33]. As transfer entropy equals mutual information in the absence of feedback, it also obeys $\mathcal{T}_{X_1 \rightarrow X_3} \leq \mathcal{T}_{X_1 \rightarrow X_2}$. This relation bounds the amount of information that can be transmitted from input to output through an intermediate variable, yet is only valid in the absence of feedback. In the presence of an $X_2 \rightarrow X_1$ feedback, i.e., $X_1 \rightleftharpoons X_2 \rightarrow X_3$, the mutual information continues to obey its DPI since X_1 and X_3 remain independent conditional on X_2 (see SM [22] Sec. SM-I), while the transfer entropy formally does not [34,35]. This can be rationalized by considering the limit $X_1 \leftarrow X_2 \rightarrow X_3$, where X_2 controls both X_1 and X_3 . Here, the X_1 trajectories would still be predictive of fluctuations in X_3 ;

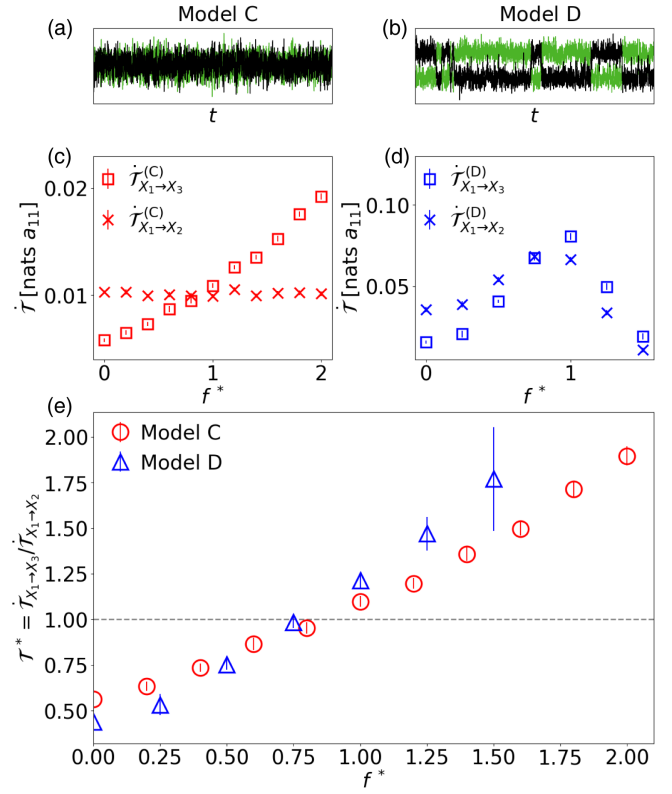


FIG. 3. (a) and (b) are typical X_1 (green) and X_2 (black) trajectories for models C and D at $a_{12} = 2a_{21} = 0.4$ and $a_{12} = a_{21} = -6$, respectively. (c) Transfer entropy rates in model C, denoted as $\dot{T}_{X_1 \rightarrow X_3}^{(C)}$ and $\dot{T}_{X_1 \rightarrow X_2}^{(C)}$, as a function of an increasing ratio of feedback to feedforward strength $f^* = a_{12}/a_{21}$, where the feedforward strength is kept constant at $a_{21} = 0.2$. (d) Similar to (c) but for model D, where the feedforward strength is kept constant at $a_{21} = -4$. (e) Ratio of transfer entropy rates overcomes DPI bound (dashed line at $\mathcal{T}^* = 1$) with increasing feedback.

thus $\mathcal{T}_{X_1 \rightarrow X_3} \geq 0$ even as $\mathcal{T}_{X_1 \rightarrow X_2} = 0$. However, where the crossover from the feedforward to the feedback-dominated regime occurs, and to what extent transfer entropies can overcome the DPI in practice, is currently not understood.

We implement the motif $X_1 \rightleftharpoons X_2 \rightarrow X_3$ in two diffusive models of mutual repression between X_1 and X_2 , labeled as models C and D. In both models, X_3 rapidly copies the state of X_2 such that the information loss from X_2 to X_3 is low. Model C is a three-dimensional OU process with linear feedback, where the ratio of the feedback to feedforward spring constants $f^* \equiv a_{12}/a_{21}$ is varied to study the violation of DPI. Model D is a nonlinear extension of C inspired by a genetic toggle switch [36] where the drifts for X_1 and X_2 are changed to $F_1 = -a_{11}X_1 - a_{12}(1 + X_1^2)/(1 + X_1^2 + X_2^2)$ and $F_2 = -a_{22}X_2 - a_{21}(1 + X_2^2)/(1 + X_1^2 + X_2^2)$; we choose $a_{12}/a_{21} = f^*$ such that the two models can be compared. Other model parameters are provided in SM [22] Sec. SM-A. Typical X_1 and X_2 trajectories at high feedback are shown in

TABLE I. Systematic errors and computational costs of approximate methods in the linear model A and the nonlinear model D. The input data size has been kept comparable whenever possible within nonprohibitive computational costs.

		Exact	Gaussian	KSG ($k \rightarrow \infty$)	KSG ($k = 0$)
Linear model (model A)	Input data size NM_1	1.6×10^6	1.6×10^6	1.6×10^5	1.6×10^6
	$\dot{T}_{1 \rightarrow 2}$ [10^{-2} nats a_{11}]	3.98 ± 0.06	3.90 ± 0.33	5.82 ± 0.07	8.98 ± 3.39
	CPU hours $\mathcal{C}$	1.88	1.35	75.92	9.55
Nonlinear model (model D)	Input data size NM_1	1.6×10^6	1.6×10^6	1.28×10^5	1.6×10^6
	$\dot{T}_{1 \rightarrow 3}$ [10^{-2} nats a_{11}]	4.23 ± 0.10	5.33 ± 0.46	2.25 ± 0.04	2.61 ± 3.48
	CPU hours $\mathcal{C}$	1.90	19.00	71.50	9.13

Figs. 3(a) and 3(b). Model C merely shows regression to the mean for both X_1 and X_2 , while model D additionally shows switching between (low, high) and (high, low) values of (X_1, X_2) in the regime of both strong feedforward and feedback coupling, around $f^* = 1$.

We plot in Figs. 3(c) and 3(d) transfer entropy rates for the two models. Each data point is obtained from steady state trajectories using TE-PWS. In model C, increasing the feedback strength f^* keeps $\dot{T}_{X_1 \rightarrow X_2}$ unchanged while $\dot{T}_{X_1 \rightarrow X_3}$ monotonically increases. The former follows analytically from the expression for the transfer entropy in a two-dimensional OU process [27] (see EM). The latter is an empirical result that arises because, when f^* is increased, X_1 becomes increasingly correlated with X_2 while X_2 continues to be copied accurately by X_3 . In contrast, in model D, both $\dot{T}_{X_1 \rightarrow X_2}$ and $\dot{T}_{X_1 \rightarrow X_3}$ peak near the switching regime, phenomenologically similar to how a bistable three-node motif with feedback has been shown to behave through approximate theory [15]. The values of the transfer entropy rates in model D are also amplified many-fold compared to model C due to stronger correlations between X_1 and X_2 resulting from the stronger, nonlinear coupling, with the concomitant switching.

In Fig. 3(e) we have plotted the degree of DPI violation in both models, quantified as the ratio of transfer entropy rates $\mathcal{T}^* \equiv \dot{T}_{X_1 \rightarrow X_3} / \dot{T}_{X_1 \rightarrow X_2}$, as a function of the ratio of the feedback to feedforward strengths, f^* . Surprisingly, we find that regardless of the nature of the variation of the individual transfer entropy rates with increasing feedback, i.e., monotonic or nonmonotonic [Figs. 3(c) and 3(d)], the ratio $\mathcal{T}^*$ monotonically increases with increasing feedback [Fig. 3(e)]. Moreover, the ratio overcomes the DPI bound when the strength of the feedback becomes comparable to that of the feedforward coupling. Our results thus show that when the feedback $X_2 \rightarrow X_1$ dominates over the feedforward interaction, $X_1 \rightarrow X_2$, the feedforward entropy $\mathcal{T}_{X_1 \rightarrow X_3}$ becomes larger than the feedforward entropy $\mathcal{T}_{X_1 \rightarrow X_2}$. The mutual information, on the other hand, continues to obey its DPI, $I(X_{1,[0,N]}; X_{3,[0,N]}) \leq I(X_{1,[0,N]}; X_{2,[0,N]})$ (see SM [22] Sec. SM-I). We expect further analytical work in the OU process to be able to support this empirical result. Moreover, as the transfer

entropy has been shown to limit the costs and benefits of information transmission [6,7], we expect the violation of the DPI to have significant physical consequences in functional networks.

Accuracy of approximate methods—An exact method is essential for the rigorous testing of the accuracy of approximate methods. This is now possible for the first time with our technique. We have compared the ground-truth TE-PWS results with those obtained from the Gaussian framework and the KSG algorithm [37] in the linear model A and in the nonlinear model D in Table I. We find that all approximate methods incur large systematic errors depending on the model. The Gaussian framework as expected gives accurate results in the linear model but fails in the nonlinear model. $\dot{T}^{[k]}$ from the KSG method in the limit ($k \rightarrow \infty$) has a systematic error due to (a) its assumption of local uniformity of the distribution of the data points and (b) the downsampling that becomes necessary to obtain a converged estimate (see SM [22] Sec. SM-H). To avoid the systematic error, the KSG algorithm is often truncated at only the $k = 0$ term of Eq. (3), yielding the so-called single-step transfer entropy rate [1,11,27,38]. As the table shows and is demonstrated in Fig. 2(a), the single-step transfer entropy rate is in general not an accurate estimate of the long-time limit. Interestingly, as the table shows, TE-PWS is not only exact but also highly cost-efficient, being either comparable or orders of magnitude cheaper than approximate methods. This is because (a) TE-PWS can calculate the transfer entropy for all k in one run, and hence can average over all k in the long time limit as shown in Eq. (2), and (b) for any given k , the cost for TE-PWS scales proportional to k , while for the Gaussian and KSG methods the scaling is, respectively, $\sim k^3$ (see SM [22] Sec. SM-H, which include Refs. [39–50]) and superlinear due to the cost of nearest neighbor search [51,52].

In conclusion, we have developed a method that for the first time makes it possible to compute transfer entropies exactly for any stochastic model. We expect transfer entropies computed by TE-PWS to be used as *ground truth* for a wide range of goals, such as the characterization and design of information flow in natural and engineered information-processing systems, and causality detection.

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Data availability—The data that support the findings of this Letter are openly available [53].

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End Matter

Analytical expression in two-dimensional OU process— In a two-dimensional OU process (X_1, X_2) with spring constants a_{ij} and diffusion constant constants D_{ij} for $i, j \in \{1, 2\}$, the steady-state transfer entropy rate $\dot{T}_{X_1 \rightarrow X_2}$ is given by [27]

$$\dot{T}_{X_1 \rightarrow X_2} = \frac{1}{2} \left(r_2 - a_{11} + \frac{D_{12}}{D_{22}} a_{21} \right), \quad (\text{A1})$$

where $r_2 = [a_{11}^2 + (D_{11}/D_{22})a_{21}^2 - 2(D_{12}/D_{22})a_{11}a_{21}]^{1/2}$. On the other hand, if Eq. (3) is truncated at the first term

after starting from a steady state, the resultant approximation, called the “single-step transfer entropy rate,” is given by [27]

$$\lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \mathcal{T}_{X_1 \rightarrow X_2}^{[0]} = \frac{1}{4D_{22}} \frac{a_{21}^2 |\mathbf{V}|}{V_{22}}, \quad (\text{A2})$$

where $\mathbf{V}$ is the stationary covariance matrix with elements $V_{ij} \equiv \langle X_i(0)X_j(0) \rangle$, analytical expressions for which can be found in [27]. In general, depending on parameter values such as feedback and noise strengths, Eqs. (A1) and (A2) yield different values. Specifically, Eqs. (A1)

and (A2) have been used to plot the two distinct dashed lines in Fig. 2(a). Additionally, we note that for model C in the section “Data processing inequality” of the main text, the dynamics of X_1 and X_2 comprises that of a two-dimensional OU process because there is no feedback from X_3 , only between X_1 and X_2 . Hence the transfer entropy rate $\dot{T}_{X_1 \rightarrow X_2}$ is given by Eq. (A1), which does not depend on $X_2 \rightarrow X_1$ feedback strength a_{12} , i.e., on f^* , as stated in the main text [see Fig. 3(c)].

Jump process—For systems with jumps between a discrete number of states and finite waiting times between the jumps, such as a well-stirred chemical reaction network described by a master equation, or a neural spiking process, the dynamics is governed by a jump propensity matrix $\mathbf{Q}$ of dimensions $\mathcal{N}^d \times \mathcal{N}^d$, which describes jumps among the $\mathcal{N}$ states of each of the d components of $\mathbf{X}$ [54]. In contrast with diffusive processes that must be time-discretized to simulate, jump processes can be simulated exactly with event-driven kinetic Monte Carlo algorithms such as the Gillespie algorithm [55]. Hence, the transfer of information from X_i to X_j during the $(k+1)$ th trajectory segment occurs at all instants of time within the segment $X_{j,[k,k+1]}$, rather than at only the endpoint $X_j(k+1)$. Therefore in the definition of the stepwise increments to transfer entropy, $\mathcal{T}_{X_i \rightarrow X_j}^{[k]}$, in Eq. (5), the logarithms of the probabilities $P(X_j(k+1)|X_{i,[0,k]}, X_{j,[0,k]})$ and $P(X_j(k+1)|X_{j,[0,k]})$ should be replaced with functionals of the entire $X_{i,[k,k+1]}$ and $X_{j,[k,k+1]}$ segments,

$$\pi_{X_i \rightarrow X_j} = - \int_{k\delta t}^{(k+1)\delta t} dt' \lambda_{ij}(t') + \sum_{\alpha=1}^{N_j} \ln \mathcal{Q}_{ij}(\alpha) \quad (\text{B1})$$

$$\pi_{X_j} = - \int_{k\delta t}^{(k+1)\delta t} dt' \lambda_j(t') + \sum_{\alpha=1}^{N_j} \ln \mathcal{Q}_j(\alpha) \quad (\text{B2})$$

$$\mathcal{T}_{X_i \rightarrow X_j}^{[k]} = \langle \pi_{X_i \rightarrow X_j} - \pi_{X_j} \rangle, \quad (\text{B3})$$

where α counts the jumps that change the state of X_j , λ_{ij} , and λ_j are escape propensities for X_j in the marginal spaces of (X_i, X_j) and (X_j) , respectively, and $\mathcal{Q}_{ij}(\alpha)$ and $\mathcal{Q}_j(\alpha)$ similarly are marginal jump propensities for the α th jump [2,15]. $\mathcal{Q}_{ij}$, λ_{ij} , and $\mathcal{Q}_j$, λ_j are obtained by marginalizing jump propensities from the full $\mathbf{X}$ space into the (X_i, X_j) and (X_j) spaces, respectively [2,15]. The marginalization can be performed with TE-PWS in a Monte Carlo fashion over conditional distributions of hidden variables, similar to the diffusive case (see SM

[22] Sec. SM-E). Prior work argues that the escape terms involving λ_{ij} and λ_j in Eqs. (B1) and (B2), respectively, can be omitted since they cancel each other on average [2]. We show however in SM [22] Sec. SM-F that the error in the transfer entropy estimate can be reduced by an order of magnitude by exploiting anticorrelated fluctuations between the escape and the jump terms.

TE-PWS algorithm—Central to TE-PWS is the computation of trajectory averages in a Monte Carlo fashion over simulated trajectories, and the availability of trajectory probabilities on the fly in the full d -dimensional space. To implement TE-PWS for a given stochastic model, we need to specify the method to simulate the model and an explicit functional form of the trajectory probability.

The diffusion processes modeled by Eq. (1) can be simulated using an Euler-Maruyama scheme with a fixed small time step. In our examples we have taken the time step to be equal to the duration of the trajectory segments for implementing the RR scheme, δt , for convenience. The propagation equation for the $(k+1)$ th step is $\mathbf{X}(k+1) = \mathbf{X}(k) + \delta t \mathbf{F}(k) + \sqrt{\delta t} \boldsymbol{\psi}(k)$, where $\boldsymbol{\psi}$ is a Gaussian random vector with zero mean and variance $\langle \psi_i(k) \psi_j(k') \rangle = 2D_{ij} \delta_{k,k'}$. Making time discrete results in an $\mathcal{O}(\delta t)$ error that can be made arbitrarily small by systematically decreasing the time step. The probability density of the change of state $\Delta \mathbf{X}(k)$ can be written analytically through Ito discretization of the Onsager-Machlup action as [20,21]

$$P(\Delta \mathbf{X}(k)) = \frac{1}{(4\pi\delta t)^{d/2} |\mathbf{D}|} \exp[-(\Delta \mathbf{X}(k) - \mathbf{F}(k)\delta t)^T \cdot \mathbf{D}^{-1}(\Delta \mathbf{X}(k) - \mathbf{F}(k)\delta t)/4\delta t], \quad (\text{C1})$$

where $|\mathbf{D}|$ is the determinant of the diffusion constant matrix. This form also holds for systems with inertia if the generalized coordinate vector $\mathbf{X}$ contains both positions and velocities [56,57].

Jump processes can be simulated with a Gillespie algorithm, which is exact, i.e., does not make a time step error [55]. The probability density of a trajectory segment $\mathbf{X}_{[t,t+\delta t]}$ is written as

$$\ln P(\mathbf{X}_{[t,t+\delta t]}) = - \int_t^{t+\delta t} dt' \lambda(t') + \sum_{\alpha=1}^{N_{\text{tot}}} \ln \mathcal{Q}_{\alpha}(t_{\alpha}), \quad (\text{C2})$$

where $\lambda(t')$ is the escape propensity from state $\mathbf{X}(t')$, given by the sum of the jump propensities $\mathcal{Q}_{\beta}(t')$ taking the system out of the state $\mathbf{X}(t')$ at time t' , and α sums over all jumps in the full trajectory, N_{tot} in number, occurring at times t_{α} .

The elementary steps of the TE-PWS algorithm are the same for both diffusive and jump processes. In brief, Step 1 is propagate M_1 trajectories of X_i and X_j in the full d -dimensional space.

Steps 2–5 are for computing the numerator in Eq. (6) for diffusive processes and the first term in Eq. (B3) for jump processes.

Step 2: For each pair of $(X_i^{(\nu)}, X_j^{(\nu)})$ trajectories, propagate M_2 trajectories of hidden variables X_l using the chosen reference distribution $P_0(X_{l,[0,k]})$. Initial conditions should be sampled from the same joint distribution $P(X_i(0), X_j(0), X_l(0))$ that $X_i^{(\nu)}(0)$ and $X_j^{(\nu)}(0)$ were drawn from.

Step 3: After every δt time, recalculate logarithmic weights $w^{(\mu)}$ for the trajectories defined as the logarithm of the ratio between the joint distribution $P(X_{i,[0,k]}^{(\nu)}, X_{j,[0,k]}^{(\nu)}, X_{l,[0,k]})$ and the reference distribution $P_0(X_{l,[0,k]})$, using Eqs. (C1) and (C2). This ratio is proportional to the ratio between the conditional distribution $P(X_{l,[0,k]}^{(\nu)} | X_{i,[0,k]}^{(\nu)}, X_{j,[0,k]}^{(\nu)})$ and the reference distribution $P_0(X_{l,[0,k]})$, as proven in SM [22] Sec. SM-C. Then calculate the uniformity in the weights with a uniformity parameter $\kappa = (\sum_{\mu} \exp w^{(\mu)})^2 / \sum_{\mu} \exp(2w^{(\mu)})$, where sums of exponentials of weights are always performed with the Log-Sum-Exp trick [58].

Step 4: Calculate the contribution to the transfer entropy in the space of (X_i, X_j) at the $(k+1)$ th step, denoted as $\mathcal{T}_a^{(\nu)}[k]$. This is done by computing the average in Eq. (9) for diffusive processes, and taking an expectation of

Eq. (B1) for jump processes (see SM [22] Sec. SM-E).

Step 5: If $\kappa < M_2/2$, resample the M_2 trajectories with the accumulated weights $w^{(\mu)}$ [see Fig. 1(c)]. This means we sample M_2 trajectories randomly with weights $w^{(\mu)}$ from the simulated trajectory ensemble, with replacement. Set all weights to zero after every resampling.

Step 6: Step 6 is for computing the denominator in Eq. (6) for diffusive processes and the second term in Eq. (B3) for jump processes. Akin to steps 2–5, for each $(X_j^{(\nu)})$ trajectory, propagate M_2 trajectories of (X_i, X_l) using reference distribution $P_0(X_{i,[0,k]}, X_{l,[0,k]})$; accumulate weights for each trajectory as the ratio between the joint distribution $P(X_{i,[0,k]}, X_{j,[0,k]}^{(\nu)}, X_{l,[0,k]})$ and the reference distribution $P_0(X_{i,[0,k]}, X_{l,[0,k]})$; compute the contribution to the transfer entropy in the (X_j) space at the $(k+1)$ th step, denoted as $\mathcal{T}_b^{(\nu)}[k]$ [Eqs. (8) and (B2)]; and resample if the uniformity in the weights is low, i.e., $\kappa < M_2/2$.

Step 7: Finally, compute $\mathcal{T}_{X_i \rightarrow X_j}$ by combining $\mathcal{T}_a^{(\nu)}$ and $\mathcal{T}_b^{(\nu)}$ from all time steps using Eqs. (6) and (B3) for diffusive and jump processes, respectively.

For clarity, a pseudocode to compute the transfer entropy $\mathcal{T}_{X_1 \rightarrow X_3}$ in a three-variable process is given in SM [22] Sec. SM-B. We note that TE-PWS is easily parallelized as it is a Monte Carlo algorithm. Specifically, steps 1–7 can be executed independently for each of the M_1 trajectories mentioned in step 1, using parallel processors. The transfer entropy is computed at the end as an average over the results from all processors.